Zoher Kachwala

PhD Candidate in LLM Reasoning & Evaluation

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Education

May 2026 **PhD in Computer Science**, *Indiana University*, GPA: 3.8/4.0

Advisor: Filippo Menczer. Dissertation: Evaluating, Guiding, and Modeling AI for Pluralistic Online Communities

May 2026 **MS in Data Science**, *Indiana University*

May 2019 **MS in Computer Science**, *Indiana University*

May 2017 **B.Tech in Computer Engineering**, *NMIMS University*

Research Interests

My research studies how language models reason and where that reasoning breaks down. I develop methods to steer model reasoning without training (Prefill-Guided Thinking), benchmarks that expose reasoning failures at scale (PLURULE), and distillation techniques that teach smaller models to generalize their reasoning to unseen tasks (DisCloze). My current work investigates automating reasoning interventions through prefill optimization and understanding reasoning quality from model internals.

Publications

ACL 2026 **PLURULE: Detecting Rule Violations on Pluralistic Social Media**

Kachwala, Truong, Muralidharan, Kwak, An, Menczer. Multimodal, multilingual benchmark spanning 17,313 rule violations across 2,419 Reddit communities with 3,692 rules in 10 languages, processed from 15B Reddit comments. Even GPT-5.2 with extended reasoning barely outperforms a trivial baseline, demonstrating that scaling model size does not help when reasoning must adapt to different rules for every instance.

NeurIPS 2025 WS; Under Review ACL ARR **Prefill-Guided Thinking (PGT)**

Kachwala, Singh, Yang, Menczer. Novel zero-shot method that steers VLM reasoning via response prefilling. Achieves 24% F1 improvement in detecting AI-generated images across 16 image generators. Reveals a Dunning-Kruger-like effect: models converge on answers before examining evidence, and prefills counteract this by forcing the model to reason first.

In Preparation **DisCloze: Distilling Cloze Reasoning for Pluralistic Moderation**

Kachwala, An, Kwak, Menczer. Distillation framework using task-aligned prefills to generate structured reasoning traces for student models. Student models up to 32B parameters detect rule violations in 100+ unseen Reddit communities, showing that structured traces help models generalize more broadly than generic ones.

NAACL 2024 **REMATCH: Robust and Efficient Matching of Local Knowledge Graphs**

Kachwala, An, Kwak, Menczer. Evaluation metric for comparing AMR graphs achieving 5x speedup over existing methods while ranking first in semantic similarity.

ICWSM 2023 **MEIU22: Multi-Platform Election Dataset**

Aiyappa, DeVerna, ..., ***Kachwala***, ..., Menczer, Yang. 30 million posts across five platforms with open-source collection pipeline.

Current Research Projects

Active **Detecting "Aha" Moments in Chain-of-Thought**

Tracking answer-token logits during reasoning to identify when models introduce new evidence or resolve contradictions. Preliminary findings show logit spikes correlate with semantically meaningful reasoning steps, providing a training-free signal for reasoning quality.

Active **Automated Prefill Optimization**

Comparing prefill vs prompt optimization using genetic-Pareto frameworks (GEPA). Testing whether prefills reach optimal task accuracy in fewer iterations than prompts because they constrain reasoning directly.

Teaching & Mentoring

2019–2025 **Teaching Assistant**, *Indiana University*

Network Science, Artificial Intelligence, Machine Learning. Led lab sections, designed assignments and exams for 300+ students.

2025 **Research Mentoring**

Mentored a high school student and an undergraduate researcher, both became co-authors on PGT.

Technical Skills

Research	Benchmark Development, LLM-as-a-judge Evaluation, Distillation
Frameworks	**PyTorch**, Hugging Face Transformers, vLLM, DeepSpeed, TensorFlow
Languages	Python, C++, CUDA, SQL, Bash
Infrastructure	Multi-GPU Training, GPU Clusters, Distributed Computing, Ray
MLOps	Weights & Biases, Weave
Cloud	AWS, GCP

Grants & Fellowships

NSF Jetstream2 Compute Allocations

Two allocations totaling $321,100 for large-scale knowledge graph matching (REMATCH) and fine-tuning models up to 32B parameters on dedicated 4x H100 80GB server (DisCloze).

Humane Studies Fellowship, *Institute for Humane Studies $5,000*

NSF Research Traineeship (NRT), *Indiana University / NSF $5,000*

Service

Reviewer	PeerJ Computer Science
Writing	The Inexplicable Efficacy of Language Models, ACM XRDS Magazine (2023)